TRUE NORTH PARTNERS

STRUCTURAL BALANCE-SHEET FORECASTING

THE SOBOLEV STACK

Economic structure where it matters. Learned adaptation where it does not know.

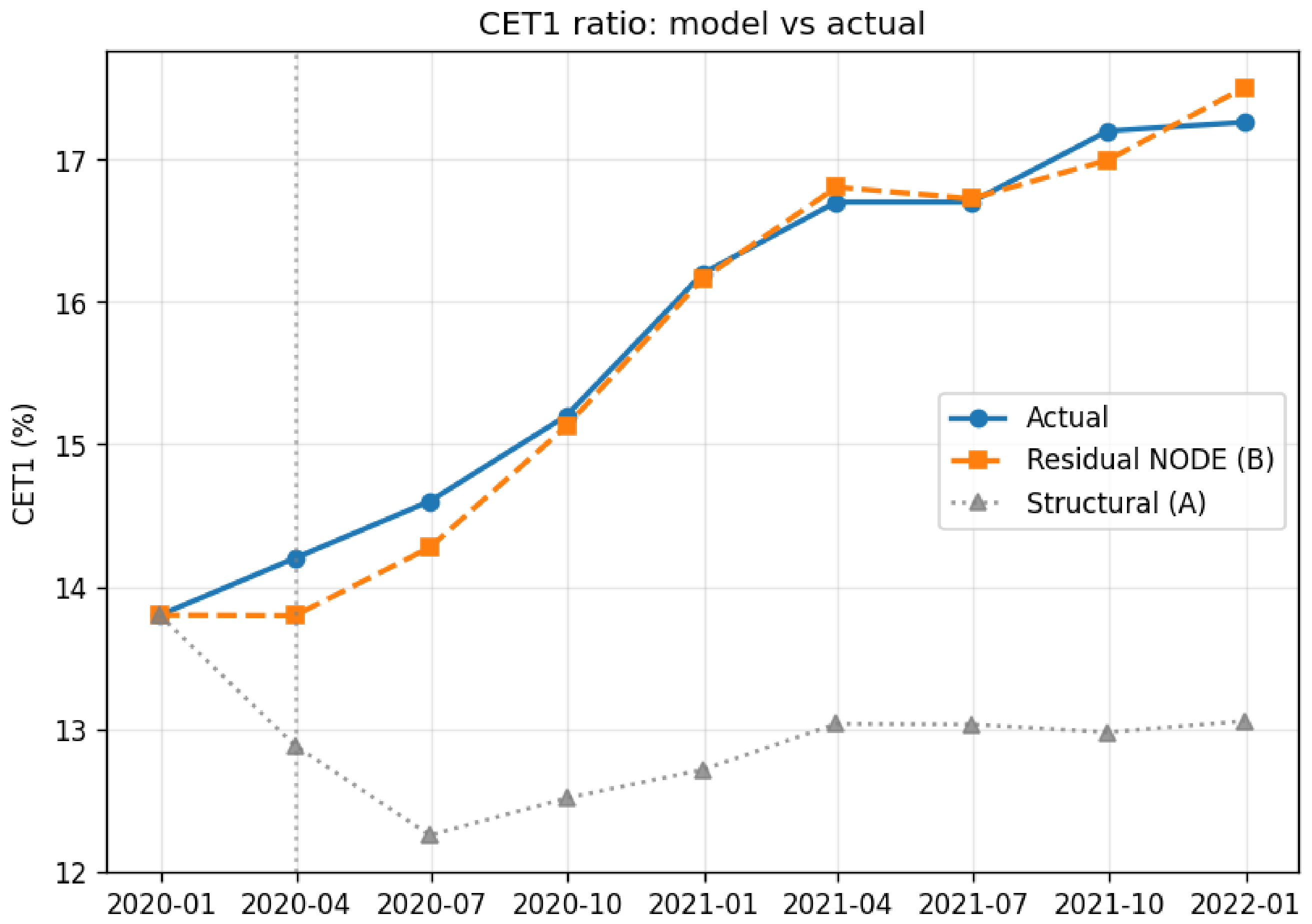
JDKF → Structural core → Residual NODE

CET1 forecasting from a single 2019 Q4 anchor

Usama Nadeem | 20 July 2026 | Internal technical seminar

THE RESULT

One anchor. Eight quarters. A decomposable CET1 forecast.



READ THIS

B follows actual CET1 through the forecast window.

A remains visible as the structural prior.

The difference is learned institution-specific behaviour, not a hidden replacement model.

The combined model tracks realised CET1 out of sample while preserving a visible structural decomposition.

Accuracy is only one part of the claim.

01

ONE ANCHOR

No quarterly re-anchoring. The trajectory is propagated from 2019 Q4.

02

OUT OF SAMPLE

The forecast period lies to the right of the declared cut-off.

03

DECOMPOSABLE

Economics and institution-specific behaviour remain separately visible.

The claim is not merely fit. It is forecast discipline plus interpretability.

Most forecasting systems optimise one virtue by sacrificing another.

FLEXIBILITY

Black-box models

Adapt well, but obscure mechanisms, weaken auditability and can extrapolate unpredictably.

CONTROL

Rigid structural systems

Remain legible, but struggle to absorb institution-specific policy and regime-dependent behaviour.

The target is the intersection.

- **We need economic structure, entity adaptation and end-to-end differentiability at the same time.**

Three questions every layer must answer.

01

COHERENCE

Do the macro variables and balance-sheet quantities tell one internally consistent story?

02

DIFFERENTIABILITY

Can outputs be calibrated and differentiated all the way back to their drivers?

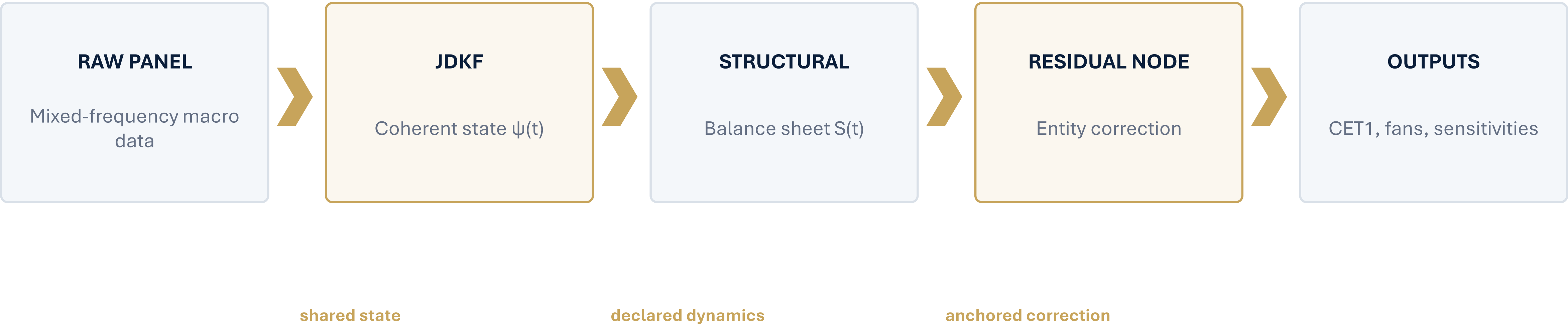
03

CONTROL

Can assumptions, identities and management actions be inspected and changed?

These are not presentation themes. They are architectural acceptance criteria.

One chain, with no broken links.



- **Each layer adds capability without discarding the properties inherited from the previous layer.**

LAYER 01

THE MACRO STATE

The downstream model needs a coherent path, not a collection of independent forecasts.

Raw macro data are measurements of a smaller shared system.

Different calendars, revisions and noise are observations of common latent forces.

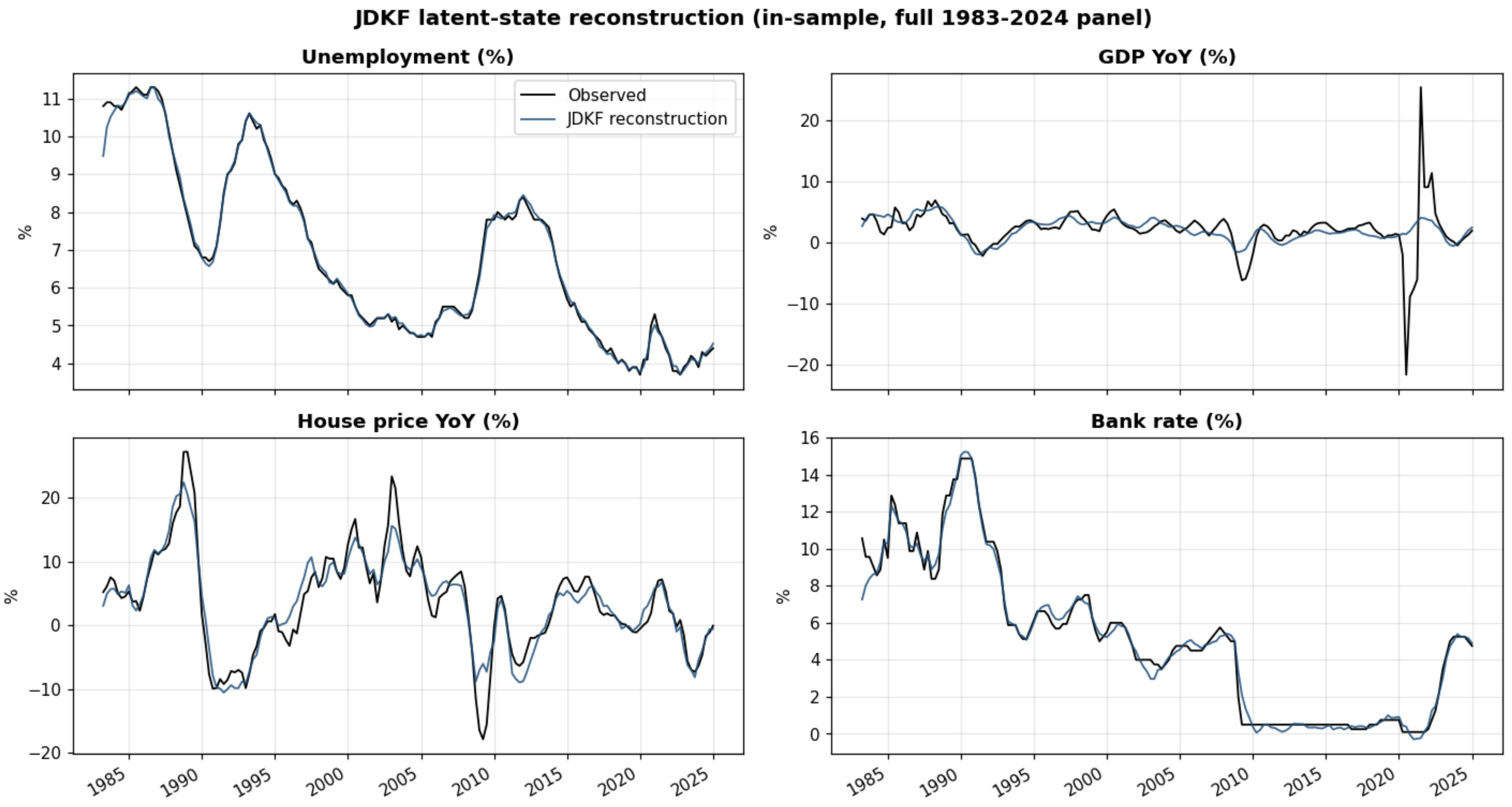
$$\psi_{t+1} = A\psi_t + w_t, \qquad z_{t+1} = H\psi_t + v_t$$

A small latent state drives every observed series through the same observation map.

- **Quarterly GDP**
- **Monthly unemployment**
- **Daily rates**
- **House prices**
- Missing and revised data

Coherence is created in the representation, not repaired after independent forecasts are produced.

The latent state reconstructs cycles without reproducing every disturbance.



- The reconstruction preserves the credit-relevant cycles while suppressing high-frequency measurement noise.

One latent object supports three downstream operations.

01

CONDITION

Fix a regulator-defined path for selected variables; infer the remainder coherently.

02

SIMULATE

Propagate the transition law to obtain coherent macro ensembles and fan charts.

03

DIFFERENTIATE

Trace a downstream metric back to every period of every macro driver.

Stress scenarios, distributions and sensitivities all begin with the same state representation.

LAYER 02

THE STRUCTURAL CORE

Turn the coherent macro state into a balance sheet that can be read and challenged.

The model is a declared system of economic and accounting dynamics.

Every material quantity has a name, an equation and a place in the balance-sheet identity.

MACRO		RISK		P&L		BALANCE SHEET		CAPITAL
$\psi(t) \rightarrow$ GDP, rates, HPI	❯	PD, LGD, NPL formation	❯	NIM, impairment, tax	❯	Loans, deposits, provisions, equity	❯	CET1 and constraints

● **The output is a coherent balance sheet, not a collection of separately forecast line items.**

Structure converts economic judgement into controlled, testable model behaviour.

The structural core contributes more than a prior forecast.

01

ECONOMIC MECHANISMS

Declare how macroeconomic shocks transmit through credit risk, margins, provisions, earnings and capital. Each relationship is an explicit hypothesis that can be examined and challenged.

02

ACCOUNTING COHERENCE

Enforce balance-sheet identities, conservation relationships and metric definitions throughout the forecast path. Every scenario remains a valid financial state.

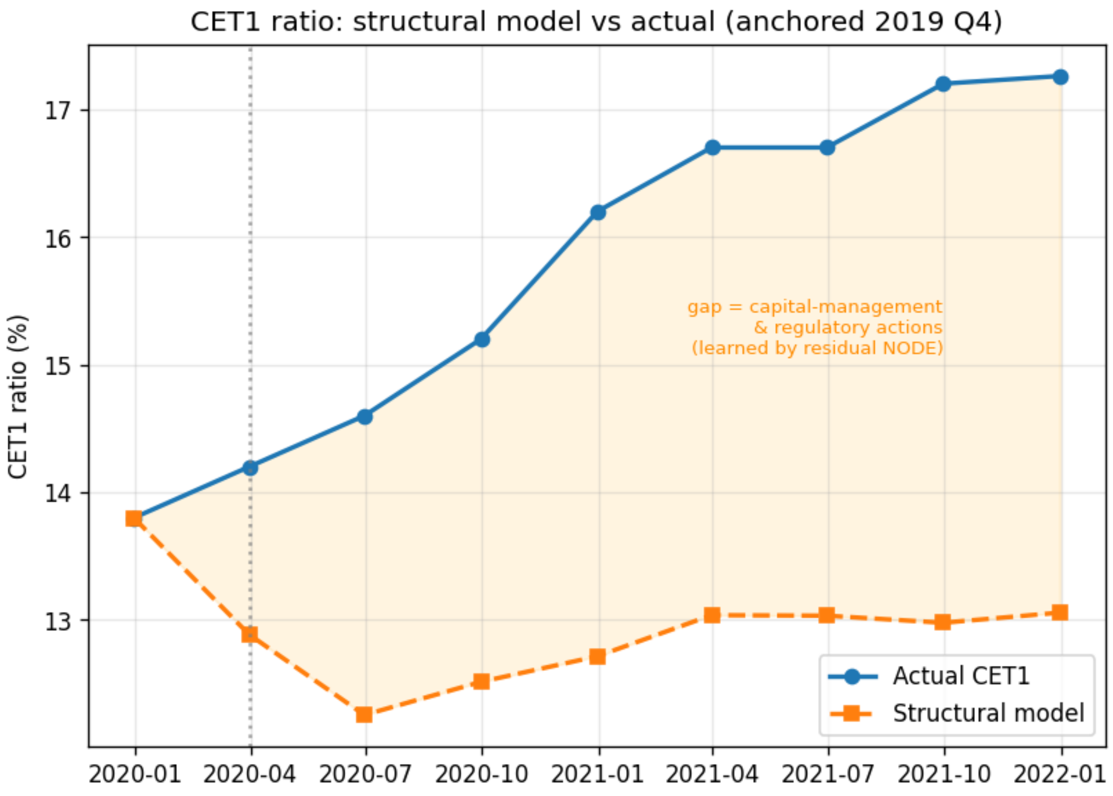
03

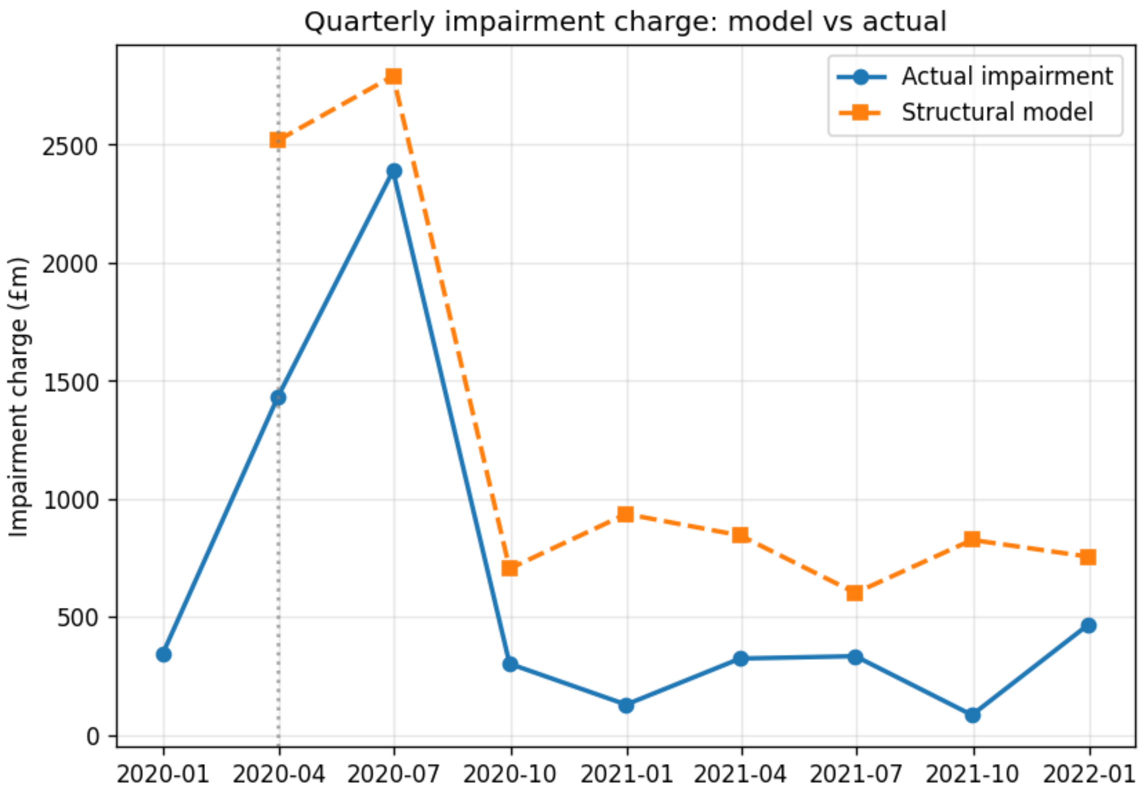
GOVERNANCE & CONTROL

Expose assumptions as named parameters and expressions. Reviewers can trace outputs, impose constraints, encode policy and alter a mechanism without retraining the entire system.

Structure makes the forecast interpretable, internally consistent and deliberately controllable before residual learning begins.

The economics are present. The institution-specific capital policy is not.





Impairment dynamics are economically plausible, yet the structural CET1 path remains below the realised path.

A structural model cannot know the institution's future decisions a priori.

The gap is where management, policy and regulation enter.

DIVIDENDS	RETENTION	BUFFERS	RELIEF
Suspension or resumption	Earnings retained in capital	Target capital management	Regulatory or transitional effects

- **The next layer should learn this gap without being allowed to erase the structural economics.**

LAYER 03

THE RESIDUAL

Learn the institution-specific correction while keeping the structural prior load-bearing.

Correction, not replacement.

$$\dot{S}_t = f_{\text{struct}}(S_t, \psi_t; \beta) + r_\theta(S_t, \psi_t)$$

STRUCTURAL DRIFT

Declared economics and identities

$$r_\theta = 0$$

recovers the structural prior exactly

RESIDUAL DRIFT

Learned entity-specific behaviour

The learner earns deviation from the prior only where the observed trajectory requires it.

The residual is fitted on the real output and pulled back toward structure.

$$L_{\text{Loss}} = \sum_t \omega_t \left\| m(S_t^{\theta}) - y_t \right\|_2^2 + \lambda \sum_t \left\| D^{-1}(S_t^{\theta} - S_t^{\text{struct}}) \right\|_2^2$$

01

DIRECT OUTPUT LOSS

Train on simulated CET1 or ECL, not a fabricated intermediate target.

02

END-TO-END GRADIENT

Differentiate through integration, metrics and accounting identities.

03

STRUCTURAL ANCHOR

Revert toward the prior where the sample does not constrain the correction.

The anchor converts structural knowledge into an explicit generalisation discipline.

Realised CET1 is separated into economics and institution-specific behaviour.

STRUCTURAL ECONOMICS

Macro transmission
Impairment
Balance-sheet mechanics
Accounting identities

+

ENTITY CORRECTION

Capital actions
Management policy
Regulatory effects
Institutional behaviour

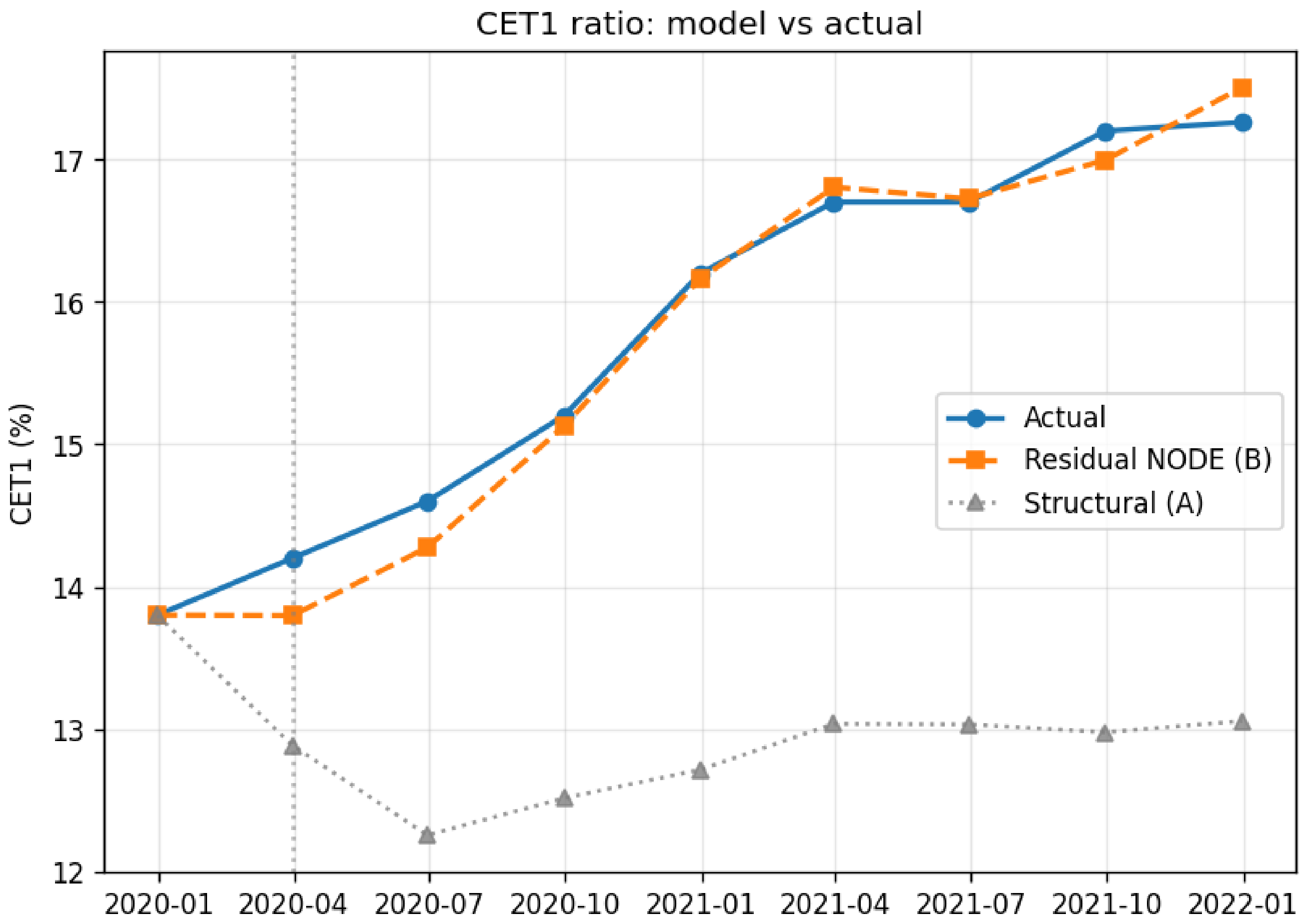
=

CALIBRATED CET1

Accurate
Auditable
Scenario-ready
Differentiable

The decomposition makes the forecast useful for governance, not merely accurate.

The correction closes the gap without hiding the prior.



WHAT TO LOOK AT

- Single anchor
- Actual remains visible
- Structural prior remains visible
- **B follows the realised path**

- **The fitted residual improves the trajectory while preserving the model's economic reference case.**

Differentiability turns a fitted forecast into an analytical engine.

SENSITIVITIES

Exact gradients by driver and horizon

FAN CHARTS

Distributional output from macro paths and residual ensembles

REVERSE STRESS

Optimise the plausible path that produces a target breach

DRILL-DOWN

Trace portfolio metrics to declared states and facilities

These capabilities exist because coherence and differentiability survive every layer.

The architecture is disciplined, not magical.

01

CONDITIONAL CLAIM

The CET1 backtest is conditional on the realised macro path.

02

DATA SUPPORT

A residual cannot learn a regime or action absent from the training sample.

03

MODEL RISK

The anchor reduces extrapolation risk; it does not remove specification risk.

Structure narrows the failure modes and makes them inspectable, but validation remains essential.

The chain holds the three acceptance criteria at every stage.

	JDKF	STRUCTURAL CORE	RESIDUAL NODE
COHERENCE	Shared state	Identities	Constrained correction
DIFFERENTIABILITY	Smooth path	Dual backend	End-to-end autograd
CONTROL	Named observables	Declared equations	Anchored deviation

Emergent capabilities: sensitivities • fan charts • reverse stress • drill-down

- No single layer delivers the proposition. The preservation of properties across layers does.

THE CLAIM

STRUCTURE EXPLAINS THE ECONOMICS.

THE RESIDUAL LEARNS THE INSTITUTION.

A coherent macro state. A transparent balance sheet. A disciplined correction.

Together, they produce a forecast that is accurate, decomposable and analytically useful.

DISCUSSION

TECHNICAL APPENDIX

Detailed construction, training and validation material for questions.

JDKF state-space estimation

$$\psi_{t+1} = A\psi_t + w_t, \qquad z_{t+1} = H\psi_t + v_t$$

The E-step runs a Kalman filter and RTS smoother. The M-step updates transition and covariance parameters from smoothed first and lag-one second moments.

Backup material for technical discussion.

ADM latent geometry

$$P_{ij} \propto \exp\left(-d\left(z_i, z_j\right)/(2\varepsilon)\right)$$

Anisotropic diffusion maps use a locally scaled Mahalanobis geometry, density normalisation and a spectral gap to select the reduced state dimension.

Backup material for technical discussion.

Structural identity preservation

$$AS_t = b, \quad A\dot{S}_t = \mathbf{0}$$

The declared drift is projected into the identity nullspace; numerical stepping is followed by bounds enforcement and identity repair.

Backup material for technical discussion.

Anchored residual objective

$$L_{\text{Loss}} = L_{\text{output}} + \lambda L_{\text{anchor}}$$

The residual is zero-initialised. Direct output loss is evaluated through the differentiable metric evaluator; the anchor regularises deviation from the structural path.

Backup material for technical discussion.

Adjoint sensitivities

$$\nabla_x F = \frac{\partial F}{\partial S}\frac{\partial S}{\partial x}$$

One reverse pass yields exact discrete-adjoint gradients with respect to every driver period, initial state and trainable structural coefficient.

Backup material for technical discussion.

Backtest discipline

$$S_{t_0} \text{ anchored}, \quad S_{t>t_0} \text{ propagated}$$

The result uses anchor-and-run rather than rolling refits. The macro path is realised, so the test isolates balance-sheet and capital-dynamics performance.

Backup material for technical discussion.